Lecture 08: Higher order linear ODE with constant coefficients (Non-homogeneous)

Rule 5: If $f(D)\,y = (x^n \sin bx \text{ or } x^n \cos bx)$; $n > 1$ and integer

$$\therefore\ y_p = \frac{1}{f(D)}\left(x^n \sin bx \text{ or } x^n \cos bx\right)$$

$$y_p = \text{I.P. of } \frac{1}{f(D)}\, x^n e^{ibx} \quad \text{(Rule 4)}$$

$$e^{ibx} = \cos bx + i \sin bx$$

$$\text{I.P. of } e^{ibx} = \sin bx$$

$$y_p = \text{R.P. of } \frac{1}{f(D)}\, x^n e^{ibx} \quad \text{(Rule 4)}$$

$$e^{ibx} = \cos bx + i \sin bx$$

$$\text{R.P. of } e^{ibx} = \cos bx$$

Example 01: Solve $(D^2+1)\,y = x^2 \cos x$ — (1)

Ans: Given $(D^2+1)\,y = x^2 \cos x$

A.E. $m^2+1=0 \Rightarrow m = \pm i$

$$y_c = A\cos x + B \sin x$$

(1) $\Rightarrow$ $$y_p = \frac{1}{D^2+1}\, x^2 \cos x$$

$$= \text{R.P. of } \frac{1}{D^2+1}\, e^{ix} x^2 \quad \text{(following Rule 4)}$$

$$= \text{R.P of } e^{ix} \frac{1}{(D+i)^2+1}\, x^2$$

$$\Rightarrow y_p = R.P.\text{ of } e^{in}\frac{1}{D^2+2Di-1+1}x^2$$

$$= R.P.\text{ of } e^{in}\frac{1}{D^2+2Di}x^2 \quad \text{(Following Rule i)}$$

$$= R.P.\text{ of } e^{in}\frac{1}{2Di\left(1+\frac{D}{2i}\right)}x^2$$

$$y_p = R.P.\text{ of } e^{in}\frac{1}{2Di}\left(1+\frac{D}{2i}\right)^{-1}x^2$$

$$= \text{"} \quad \text{"} \quad e^{in}\frac{1}{2Di}\left\{1-\frac{D}{2i}+\left(\frac{D}{2i}\right)^2-\cdots\right\}x^2$$

$$= \text{"} \quad \text{"} \quad e^{in}\frac{1}{2Di}\left(x^2-\frac{\cancel{2}x}{\cancel{2}i}+\frac{2}{-4}-0\cdots\right)$$

$$= \text{"} \quad \text{"} \quad e^{in}\frac{1}{2Di}\left(x^2-\frac{xi}{i^2}-\frac{1}{2}\right)$$

$$= \text{"} \quad \text{"} \quad e^{in}\frac{i}{2Di^2}\left(x^2+xi-\frac{1}{2}\right)$$

$$= \text{"} \quad \text{"} \quad e^{in}\frac{-i}{2D}\left(x^2+xi-\frac{1}{2}\right)$$

$$= \text{"} \quad \text{"} \quad -\frac{i}{2}(\cos x+i\sin x)\left(\frac{x^3}{3}-\frac{x^2}{2}i-\frac{1}{2}x\right)$$

$$= \text{"} \quad \text{"} \quad -\frac{i}{2}\left(\cos x\,\frac{x^3}{3}-\frac{x^2}{2}i\cos x-\frac{1}{2}x\cos x\right.$$

$$\left.-i\,\frac{x^3}{3}\sin x+\frac{x^2}{2}\sin x-\frac{1}{2}xi\sin x\right)$$

$$= \text{"} \quad \text{"} \quad -\frac{i}{2}\left\{\left(\frac{x^3}{3}\cos x-\frac{1}{2}x\cos x+\frac{x^2}{2}\sin x\right)+i\left(-\frac{x^2}{2}\cos x\right.\right.$$

$$\left.\left.-\frac{x^3}{3}\sin x-\frac{1}{2}x\sin x\right)\right.$$

$$y_p = \left(-\frac{1}{2}\,\frac{x^2}{2}\cos x-\frac{1}{2}x^3\sin x\right.$$

$$\left.-\frac{1}{2}x\sin x\right)$$

G.S: $y = y_c + y_p$

Example 02 (example 3 from the lecture 07) Solve

$$(D^2+1)y = \sin x \quad \text{--- (1)}$$

Ans: $y_p = \frac{1}{D^2+1}\sin x$

$= \text{I.P. of } \frac{1}{D^2+1} e^{ix}$ (Rule 2) failure case)

$= \text{" "} \; e^{ix} \frac{1}{(D+i)^2+1} 1$

$= \text{" "} \; e^{ix} \frac{1}{D^2+2Di-1+1} 1$

$= \text{" "} \; e^{ix} \frac{1}{2Di\left(1+\frac{D}{2i}\right)} 1$

$= \text{" "} \; e^{ix} \frac{i}{2Di^2}\left(1-\frac{D}{2i}+\cdots\right) 1$

$= \text{" "} \; e^{ix} \frac{-i}{2D}(1-0)$

$= \text{" "} \; (\cos x + i\sin x)\left(-\frac{1}{2}\right)ix$

$= \text{I.P. of } i\left(-\frac{1}{2}x\cos x - \frac{1}{2}ix\sin x\right)$

$y_p = \text{" "} \; \left(-\frac{1}{2}ix\cos x + \frac{1}{2}x\sin x\right)$

$$\boxed{y_p = -\frac{1}{2}x\cos x}$$

(1) $\Rightarrow$ $y_p = \frac{1}{D^2+1}\sin x$

$= \frac{x}{2}\int \sin x$

$= \frac{x}{2} - \cos x$

$$\boxed{y_p = -\frac{x}{2}\cos x}$$

Special case of Rule 5: If $y_p = \frac{1}{f(D)} x^n \underbrace{(\sin bx \text{ or } \cos bx)}_{V(x)}$

(Here $n = 1$)

$$y_p = \frac{1}{f(D)} x V(x) \quad (n=1)$$

$$\Rightarrow y_p = x \frac{1}{f(D)} V(x) - f'(D) \left\{ \frac{1}{f(D)} \right\}^2 V(x)$$

<u>Example 05:</u> Solve $(D^2 - 2D + 1)y = x \sin x$

<u>Ans:</u> Given $(D^2 - 2D + 1)y = x \sin x$ —— (1) (Here $n=1$)

A.E, $m^2 - 2m + 1 = 0$

$\Rightarrow (m-1)^2 = 0 \Rightarrow m = 1, 1$

$y_c = c_1 e^x + c_2 x e^x$

(1) $\Rightarrow$ $y_p = \frac{1}{D^2 - 2D + 1} \underbrace{x \sin x}_{V(x)}$

$$= x \frac{1}{D^2 - 2D + 1} \sin x - (2D - 2) \left\{ \frac{1}{D^2 - 2D + 1} \right\}^2 \sin x$$

$$= x \frac{1}{-1^2 - 2D + 1} \sin x - \frac{2D - 2}{(-1^2 - 2D + 1)^2} \sin x$$

$$= x \frac{-1}{2D} \sin x - \frac{2D - 2}{4D^2} \sin x$$

$$= -\frac{x}{2} (-\cos x) - \frac{2D - 2}{4(-1)} \sin x$$

$$= \frac{x}{2} \cos x + \frac{1}{2} (D - 1) \sin x$$

$$= \frac{x}{2} \cos x + \frac{1}{2} (\cos x - \sin x)$$

G.S

$y = y_c + y_p$

Home task: (MKC book) P-112

Example 4.13, P-122 7, 8, 9, 10, 11, 12, *14, *16,

Example 05: Solve $(D^2+1)y = x\cos x - \cos x$

Ans: we will do in the class

G.S. $y = c_1\cos x + c_2\sin x + \frac{1}{4}x\cos x - \frac{1}{2}x\sin x + \frac{1}{4}x^2\sin x$.

Home task: Zill's book P-155 exercise 4.5

5, 6, 7, 8, 54, 57, 58